Module 4 Project Rubric

RiskLens | Optimized Analyst

C4 - Model Optimization and Deployment Decisions
Explain parallelism and model-optimization options, compare compressed versions of the trained Analyst under a predeclared quality gate, and justify an accept, reject or fallback decision using measured quality, memory, latency, throughput and cost.

Evidence to prepare

- **Technique note:** Explain parallelism, training/inference optimization, distillation, quantization and merging; justify the selected route for RiskLens.

- **D4.1:** Frozen GATE with format/content drop thresholds and their units, hourly-price assumption and intended arm; baseline, candidates and rationale recorded before re-scoring.

- **D4.2:** At least two compressed versions of the trained Analyst, with measured memory footprints and compression ratios against the half-precision reference.

- **D4.3:** Frozen-set format/content results against Stage 3, declared thresholds, all pass/fail outcomes and an explicit accept/reject/fallback decision.

- **D4.4:** Per-arm p50/p95, throughput from actual generated tokens and estimated cost per 1,000 reports, with hardware, run count, warm-up and pricing assumptions.

- **Module 4 checkpoint:** Saved standalone quality and benchmark outputs for D4.1-D4.4. The existing combined Stage 4 evaluation cell runs after integration in Module 5.

How grading works

All 5 criteria must be **Met** to complete C4. **C4.2 is MUST PASS** and is reviewed first. If it is Not Yet, revise before the remaining criteria are reviewed. No points, weights or bonus credit.

Submit the same notebook link with saved outputs, a milestone snapshot, a short summary and known issues. Revise Not Yet criteria and affected evidence. The full RiskLens Grading Rubric defines review rules; course policy governs attempts and timing.

Module 4 | Assessment criteria (1/2)

C4 - Model Optimization and Deployment Decisions

RiskLens Criterion Set v2.0 | All 5 criteria must be Met. Standards are identical to the consolidated rubric.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C4.1 Your optimization choice shows understanding	Technique note beside D4.1-D4.2; links to the selected model configurations.	Distinguish data from model parallelism, training from inference optimization, teacher-output from logit distillation, adapter folding from multi-model merging, and reduced-precision quantization. Identify the RiskLens bottleneck and justify the chosen route and trade-offs. Independent multi-GPU training, distillation and multi-parent merging are optional extensions, not required implementations.	Calling device placement data-parallel training; confusing adapter folding with merging several fine-tunes; claiming compression always accelerates inference; listing terms without a choice.
C4.2 Your quality gate precedes the comparison MUST PASS	D4.1 GATE and pre-score snapshot/version; baseline and candidate inventory.	Before compressed-model re-scoring, record justified maximum format and content drops, their units, the Stage 3 reference, candidate arms, hourly price and intended ship arm. Preserve the dated version; keep thresholds unchanged for that comparison. Explain why the tolerances are acceptable for the stated use case rather than choosing them to guarantee a pass.	Setting thresholds after seeing results; confusing percent with percentage points; changing the intended arm without a separate decision record; unjustified permissive tolerances.
C4.3 Your compression comparison uses the trained Analyst	D4.2 memory table; D4.3 saved generations, quality table and Module 4 checkpoint.	Produce at least two compression arms of the learner-trained Analyst and record memory footprints and ratios against half precision. Re-score each arm and the Stage 3 reference on the same frozen rows for format and content, with consistent settings. Show threshold comparisons and every verdict. Identify content cosine similarity as a proxy, not factual correctness.	Quantizing an untuned base; counting a configuration as a measured arm; measuring format only; different rows per arm; presenting cosine similarity as a factuality guarantee.

Evidence note. Assess saved outputs and linked artifacts; graders do not need to repeat model training. A missing evaluation output does not demonstrate a result. Preserve real prompts, configurations, data versions and outputs as the work happens. Dated snapshots or version history support freezing; cell position alone does not prove chronology.

Module 4 | Assessment criteria (2/2)

C4 - Model Optimization and Deployment Decisions

RiskLens Criterion Set v2.0 | All 5 criteria must be Met. Standards are identical to the consolidated rubric.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C4.4 Your serving estimates are measured and qualified	D4.4 benchmark outputs for reference and each arm; settings and cost calculation.	Report p50/p95 latency, actual generated-token throughput and estimated cost per 1,000 reports for the reference and each arm. State hardware, run count, warm-up, input/output lengths and hourly-price/utilization assumptions. Label small-sample percentiles and cost estimates appropriately; explain why lower precision may save memory without reducing latency.	Using max_new_tokens as generated length; comparing unequal workloads without disclosure; calling notebook p95 a production SLA; treating the assumed GPU price as a verified bill.
C4.5 Your deployment decision respects the gate	D4.3 decision note; accepted/rejected arm inventory; fallback and limits.	Select a candidate that passes the declared gate or reject failed candidates and document a feasible fallback. Link the decision to quality, memory and runtime evidence. An honestly evaluated failed compression can meet this competency; it cannot be labeled production-ready. Keep the original gate record and distinguish any later candidate or fallback evaluation.	Declaring ship after a failed gate; hiding a quality drop behind a smaller file; silently changing the model; rewriting the threshold to make a candidate pass.

Module handoff. At Module 4, add a standalone checkpoint below your workspaces for the frozen-set format/content comparison, benchmarks and verdict. Save these outputs without calling analyze_product(). Run the starter's combined Stage 4 evaluation after integration in Module 5. Do not edit GIVEN functions or use stubs to satisfy the checkpoint.

Source mapping: D4.1-D4.4. RiskLens Criterion Set v2.0 governs criteria; the published course policy governs administrative rules.